

NISHA RAJPUT

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Profile

AI/ML Engineer with experience building deep learning models, LLM-powered applications, and ML pipelines. Skilled in Python, PyTorch, HuggingFace Transformers, LangChain, FastAPI, Kubernetes, and AWS.

Experience

George Mason University

Classroom Technology Assistant / AI Engineer

Fairfax, VA

May 2025 – Present

- Engineered an AI-driven ticket classification pipeline integrating GPT-5 and LangChain, reducing manual triage overhead by 35% across 40+ enterprise classrooms via zero-shot intent routing.
- Architected high-throughput ETL pipelines ingesting 10,000+ operational log entries monthly into ML-ready feature stores, enabling real-time analytics and cutting reporting overhead by 25%.
- Deployed 5+ containerized ML inference services on Kubernetes with rolling updates, accelerating release cadence by 30% and improving SLA adherence by 35%.

ExcelR Pvt. Ltd

Full Stack Developer Intern

Hyderabad, India

Aug 2023 – Aug 2024

- Integrated an OpenAI-powered NLP module into the platform to auto-classify and route 1,000+ user queries daily, decreasing support workload by 25%.
- Optimized SQL queries and relational schema across 50,000+ user interaction records, cutting query execution latency by 30% and enabling real-time performance dashboards.
- Drove analytics-informed UI/UX improvements using ReactJS and JavaScript, amplifying user engagement by 20%.

Academic Projects

AURA — Multimodal Emotion Recognition System

AI/ML Engineering

- Fine-tuned RoBERTa (GoEmotions) and Wav2Vec2 (CREMA-D) for 7-class multimodal emotion classification across textual and acoustic signals, achieving strong macro-F1 across both modalities.
- Applied α -weighted late fusion of multimodal emotion vectors feeding a policy layer that dynamically shapes GPT-4o response tone, served via FastAPI with sub-second latency.
- Orchestrated an end-to-end pipeline (Whisper ASR $\rightarrow$ text/speech ERC $\rightarrow$ fusion $\rightarrow$ policy $\rightarrow$ LLM) with a single-page web UI, demonstrating production-grade multimodal AI orchestration.

Weather Predictor — Agentic AI System

Agentic AI · LLM Tool Use

- Developed a multi-agent system using HuggingFace smolagents with a manager-agent orchestrating a search-agent for autonomous multi-step reasoning and real-time web retrieval.
- Created a custom regex-based WeatherParserTool extending the smolagents Tool interface to extract structured temperature and condition data from unstructured search output.
- Instrumented full agent observability via Arize Phoenix OpenTelemetry tracing, integrating LiteLLM (GPT-4.1-nano) as the agent backbone for transparent reasoning visualization.

SHEO — Smart Home Energy Optimizer

Optimization · AI Decision Systems

- Designed a multi-objective LP optimizer using Pyomo + GLPK to schedule appliance usage across 24 hourly intervals under a 7 kW power cap, balancing cost, carbon, and comfort objectives.
- Configured 3 user preference profiles (cost saver, eco warrior, comfort first) via weighted multi-objective tuning, delivering projected 15–30% electricity cost savings versus baseline scheduling.
- Built contract-aware tariff logic reading per-state pricing from JSON with structured output schema.

Skills

AI/ML & Deep Learning: PyTorch, HuggingFace Transformers, Scikit-learn, RoBERTa, Wav2Vec2, BERT, Fine-tuning, Transfer Learning

LLMs & Generative AI: OpenAI API (GPT-4o), LangChain, LangChain Agents, smolagents, Prompt Engineering, RAG

Cloud & DevOps: Docker, Kubernetes, Jenkins, AWS

Databases: SQL (MySQL), MongoDB, ETL Pipelines, Pandas, NumPy

Languages & Web: Python, JavaScript, HTML/CSS, ReactJS, Git

Education

George Mason University

Master of Science (M.S.), Computer Science

Aug 2024 – May 2026

Vidya Jyothi Institute of Technology

Bachelor of Technology (B.Tech), Computer Science

May 2021 – Apr 2024

Certifications

- Python Essentials (PCAP — Programming Essentials in Python) — Cisco Networking Academy
- Oracle Database Management System (DBMS) Certification — Oracle Corporation